

Practical 1 - Review of linear regression and introduction to logistic regression

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Dataset required: `mental.RData`

Introduction

The purpose of this session is to perform a full analysis of the mental activity data trial introduced earlier. We will use a series of normal error models, and compare their respective fits using F-tests.

The data are from a three-arm placebo-controlled randomised trial assessing the effects of injection with morphine and heroin on mental activity. Seventy two people were randomized to one of three arms (placebo, morphine, heroin), and mental activity was assessed by the sum of scores on seven items of a questionnaire, with a high score indicating high mental activity.

The dataset contains three variables:

- `treat` - Randomised arm: 1=placebo; 2=morphine; 3=heroin
- `prement` - Mental activity score before injection
- `'mentact` - Mental activity score 2 hours after injection with drug.

The questions that the analysis should address are as follows.

Aims

- Do the mean post-injection mental activity scores differ among groups?
- What effect does adjustment for pre-injection mental activity score have on your conclusions?
- Are the effects of the drugs different for subjects with different levels of pre-injection mental activity?

Analysis

Question 1

Read the data into R and examine. Use tables, histograms and plots to look at the distributions of the pre- and post-injection scores. Also plot the post-injection scores against the pre-injection scores, distinguishing between the three groups.

Discuss with your colleagues what conclusions can be drawn about the appropriateness of linear regression models for post-injection scores on the basis of these preliminary explorations.

SOLUTION

```
load("mental.RData")
mental$treat <- factor(mental$treat,
                      levels = c("1", "2", "3"),
                      labels = c("Placebo", "Morphine", "Heroin"))
```

There are various ways to display summary statistics in R. Here we will make use of `dplyr` and the `skimr` package for producing summaries:

```
library(dplyr)
library(skimr)
mental %>%
  group_by(treat) %>%
  skim(prement, mentact)
```

Table 1: Data summary

Name	Piped data
Number of rows	72
Number of columns	3

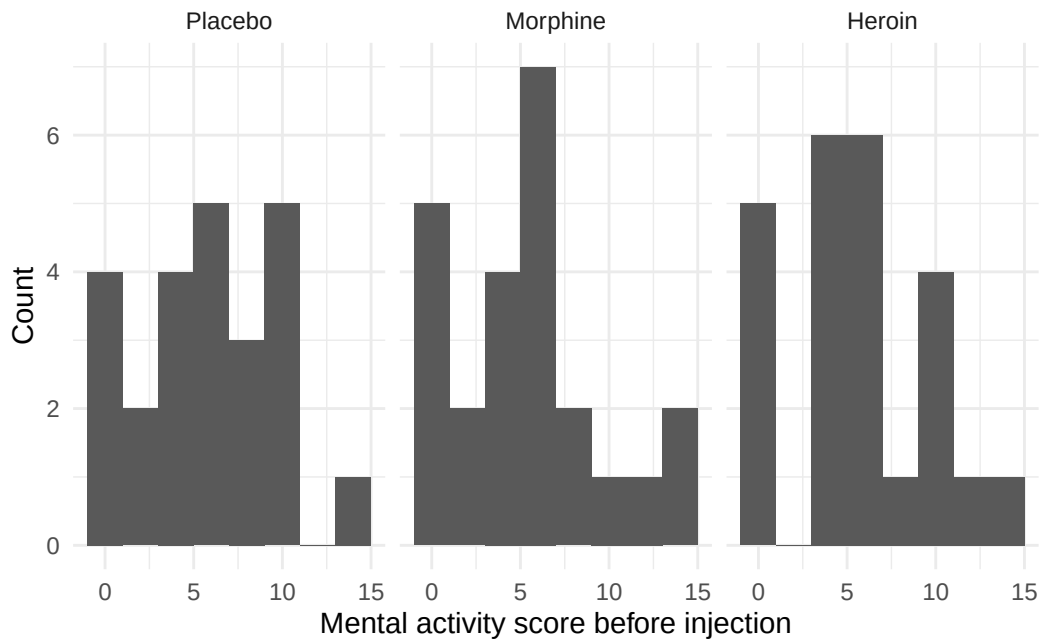
Column type frequency:	
numeric	2
Group variables	treat

Variable type: numeric

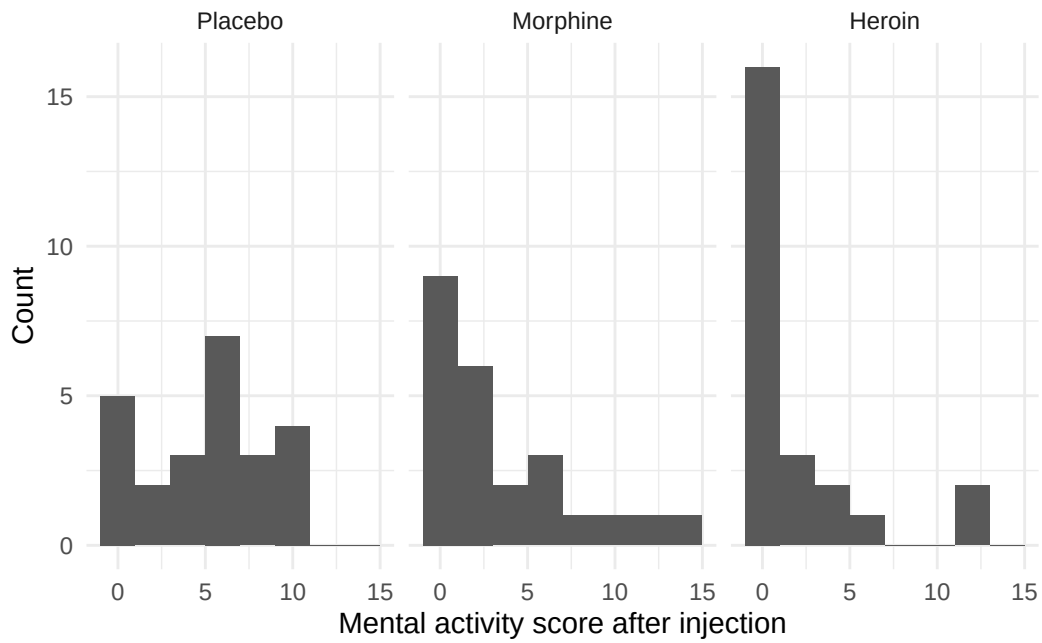
skim_variable	treat	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
prement	Placebo	0	1	6.00	3.79	0	3.75	6.0	8.50	14	
prement	Morphine	0	1	5.75	4.15	0	2.00	6.0	7.25	14	
prement	Heroin	0	1	5.88	4.08	0	4.00	6.0	9.25	14	
mentact	Placebo	0	1	5.54	3.35	0	2.75	6.0	8.00	11	
mentact	Morphine	0	1	3.67	4.12	0	0.00	2.0	6.00	14	
mentact	Heroin	0	1	2.17	3.80	0	0.00	0.5	2.00	13	

```
library(ggplot2)

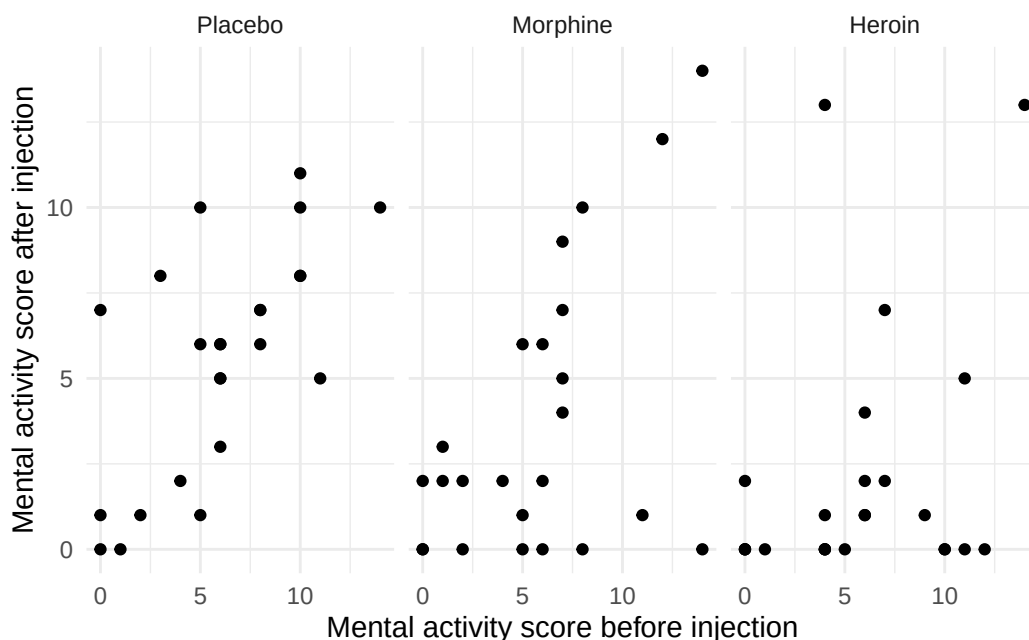
ggplot(mental, aes(x = prement)) +
  geom_histogram(binwidth = 2) +
  facet_wrap(~ treat) +
  labs(
    x = "Mental activity score before injection",
    y = "Count"
  ) +
  theme_minimal()
```



```
ggplot(mental, aes(x = mentact)) +  
  geom_histogram(binwidth=2) +  
  facet_wrap(~ treat) +  
  labs(  
    x = "Mental activity score after injection",  
    y = "Count"  
  ) +  
  theme_minimal()
```



```
ggplot(mental, aes(x = prement,  
                  y = mentact)) +  
  geom_point() +  
  facet_wrap(~ treat) +  
  labs(  
    x = "Mental activity score before injection",  
    y = "Mental activity score after injection"  
  ) +  
  theme_minimal()
```



The tabulations and plots indicate that the distributions of scores are similar pre-injection (as we would expect since this is a randomised study), but markedly different post-injection, although with small numbers it is difficult to make definitive judgements.

The histograms of pre- and post-injection scores suggest that these variables are not normally distributed. Note, however, that this does not necessarily invalidate inferences that we want to draw from the regression models that we are going to fit, since in a regression model it is only residuals that need to be normally distributed for valid inferences.

The scatter plot shows that pre- and post-injection scores are positively associated. An assumption that the relationship between the two is linear in each group does not seem unreasonable.

Question 2

Write down in algebraic form the linear predictors that correspond to each of the models described in Table 3. Carefully define each term that you use in the various models.

Note that the distributional assumption in each model is $Y_i \sim N(\mu_i, \sigma^2)$ where Y_i is the post-injection mental activity score. In each model the link function is the identity $\eta_i = \mu_i$.

Table 3: Model linear predictors

#	Terms fitted	Linear predictor
1	Overall mean	
2	Drugs	

#	Terms fitted	Linear predictor
3	Pre-inj (Pre)	
4	Drugs + Pre	
5	Drugs + Pre + (Drugs-by-Pre interaction)	

SOLUTION

Table 4: Model linear predictors completed

#	Linear predictor
1	$\eta_i = \beta_0$
2	$\eta_i = \beta_0 + \beta_1 u_i + \beta_2 v_i$
3	$\eta_i = \beta_0 + \beta_3 x_i$
4	$\eta_i = \beta_0 + \beta_1 u_i + \beta_2 v_i + \beta_3 x_i$
5	$\eta_i = \beta_0 + \beta_1 u_i + \beta_2 v_i + \beta_3 x_i + \beta_4 u_i x_i + \beta_5 v_i x_i$

Here x_i and Y_i are respectively the pre- and post-injection scores for individual i . The additional predictor variables are defined as follows:

$$u_i = \begin{cases} 0 & \text{if individual } i \text{ is in the Placebo or Heroin group} \\ 1 & \text{if individual } i \text{ is in the Morphine group} \end{cases}$$

$$v_i = \begin{cases} 0 & \text{if individual } i \text{ is in the Placebo or Morphine group} \\ 1 & \text{if individual } i \text{ is in the Heroin group} \end{cases}$$

Note that the use of the same parameter symbol, e.g. β_1 in multiple linear predictors does not mean that there is a single common value of this parameter across the models. That is, β_1 in model 2 will not take the same value as β_1 in models 4 and 5.

In each model the link function is the identity,

$$\eta_i = \mu_i,$$

and it is assumed that

$$Y_i \sim N(\mu_i, \sigma^2)$$

and the observations are independent.

Question 3

Fit and save each linear regression model using `lm`. Then use the `anova` function to complete Table 5.

Table 5: Residual sum of squares and degrees of freedom for fitted models

#	Terms fitted	RSS	Residual df
1	Overall mean		
2	Drugs		
3	Pre-inj (Pre)		
4	Drugs + Pre		
5	Drugs + Pre + (Drugs-by-Pre interaction)		

SOLUTION

```
#model 1
mod1 <- lm(mentact~1, data=mental)
mod2 <- lm(mentact~treat, data=mental)
mod3 <- lm(mentact~prement, data=mental)
mod4 <- lm(mentact~treat+prement, data=mental)
mod5 <- lm(mentact~treat*prement, data=mental)
#here we just show output for model 5:
summary(mod5)
```

Call:

```
lm(formula = mentact ~ treat * prement, data = mental)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.8281	-1.9351	-0.5161	1.4161	11.3601

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.97803	1.29407	1.529	0.13116
treatMorphine	-1.21174	1.75034	-0.692	0.49118
treatHeroin	-1.46197	1.77186	-0.825	0.41228
prement	0.59394	0.18347	3.237	0.00189 **
treatMorphine:prement	-0.08953	0.24835	-0.360	0.71963
treatHeroin:prement	-0.31299	0.25038	-1.250	0.21570

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.333 on 66 degrees of freedom

Multiple R-squared: 0.3442, Adjusted R-squared: 0.2945

F-statistic: 6.927 on 5 and 66 DF, p-value: 2.974e-05

`anova(mod5)`

Analysis of Variance Table

Response: mentact

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treat	2	137.25	68.625	6.178	0.003472 **
prement	1	228.57	228.570	20.577	2.481e-05 ***
treat:prement	2	18.93	9.464	0.852	0.431203
Residuals	66	733.13	11.108		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The table below shows the number of parameters, the residual sums of squares, and the degrees of freedoms for each of the five models (in red above).

Table 6: Model comparison based on RSS and residual degrees of freedom

#	Linear predictor	No. pars	RSS	rdf
1	$\eta_i = \beta_0$	1	1117.9	71
2	$\eta_i = \beta_0 + \beta_1 u_i + \beta_2 v_i$	3	980.6	69
3	$\eta_i = \beta_0 + \beta_3 x_i$	2	884.3	70
4	$\eta_i = \beta_0 + \beta_1 u_i + \beta_2 v_i + \beta_3 x_i$	4	752.1	68
5	$\eta_i = \beta_0 + \beta_1 u_i + \beta_2 v_i + \beta_3 x_i + \beta_4 u_i x_i + \beta_5 v_i x_i$	6	733.1	66

Question 4

Calculate the fitted values for model 5. Display these for each trial arm in a graph. Compute the intercepts and slopes of the three fitted lines from the parameters estimated in the model.

Table 7: Fitted intercepts and slopes by trial arm

Trial arm	Fitted intercept	Fitted slope
Placebo		
Morphine		
Heroin		

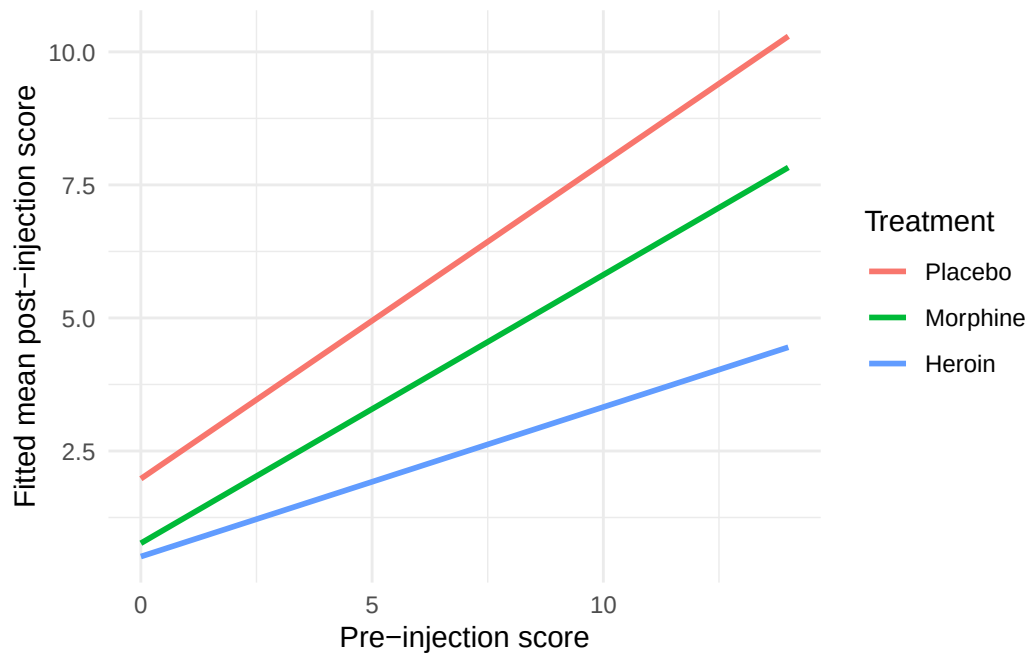
SOLUTION

This is one way of creating and plotting the fitted values:

```
#create data frame of predicted values
newdat <- expand.grid(
  prement = seq(min(mental$prement, na.rm = TRUE),
                max(mental$prement, na.rm = TRUE),
                length.out = 100),
  treat = levels(mental$treat) # assumes treat is a factor
)
newdat$fitted <- predict(mod5, newdata = newdat)

#plot them
library(ggplot2)

ggplot(newdat, aes(x = prement, y = fitted, colour = treat)) +
  geom_line(linewidth = 1) +
  labs(
    x = "Pre-injection score",
    y = "Fitted mean post-injection score",
    colour = "Treatment"
  ) +
  theme_minimal()
```



The fitted values from model 5 lie along three straight lines, one for each trial arm, with the slopes and intercepts being derived from the output as follows.

Table 8: Fitted intercepts and slopes by trial arm

Trial arm	Fitted intercept	Fitted slope
Placebo	1.97803	0.5939394
Morphine	$1.97803 - 1.211742 = 0.7662884$	$0.5939394 - 0.0895258 = 0.5044136$
Heroin	$1.97803 - 1.461968 = 0.5160623$	$0.5939394 - 0.3129855 = 0.2809539$

Question 5

Pen & paper exercise: use the results in Table 5 to compute test statistics to compare

- the fits of models 3 and 4 and
- the fits of models 4 and 5.

In each case convert the test statistic to a p-value using statistical tables (good practice for the summer examination!) or the `pf` function.

Confirm your results using the `anova` command.

Discuss: What do you conclude from these tests? Why can an analogous test not be used to compare the fits of models 2 and 3?

SOLUTION

To compare the fits of models 3 and 4 compute the relevant F-statistic from Table 6:

$$F = \frac{(884.3 - 752.1)/(70 - 68)}{752.1/68} = 5.98$$

Comparison with the $F_{2,68}$ distribution (`pf(5.98, 2, 68, lower.tail=FALSE)`) gives $p = 0.004$ ($0.001 < p < 0.005$ if using statistical tables). There is therefore strong evidence of a difference in the post-injection scores between the groups conditional on (holding constant) pre-injection scores.

In R we can do this using:

```
anova(mod3, mod4)
```

Analysis of Variance Table

Model 1: mentact ~ prement

Model 2: mentact ~ treat + prement

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	70	884.33				
2	68	752.06	2	132.27	5.98	0.004052 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

To compare the fits of models 4 and 5 we have:

$$F = \frac{(752.1 - 733.1)/(68 - 66)}{733.1/66} = 0.85$$

Comparison with the $F_{2,66}$ distribution (`pf(0.85, 2, 66, lower.tail=F)`) gives $p = 0.43$ ($p > 0.1$ from statistical tables). There is therefore no evidence of an interaction between group and pre-injection score. In R:

```
anova(mod4, mod5)
```

Analysis of Variance Table

Model 1: mentact ~ treat + prement

```
Model 2: mentact ~ treat * prement
      Res.Df  RSS Df Sum of Sq    F Pr(>F)
1       68 752.06
2       66 733.13  2    18.928 0.852 0.4312
```

Models 2 and 3 cannot be compared using an F-test because the two models are not nested.

Question 6

Use results from models 2 and 4 to complete Table 9.

Table 9: Means and differences from placebo

Trial arm	Mean	Mean difference from placebo	SE	Adjusted difference from placebo	SE
Placebo		0	–	0	–
Morphine					
Heroin					

SOLUTION

Trial arm	Mean	Mean difference from placebo	SE	Adjusted difference from placebo	SE
Placebo	5.542	0	–	0	–
Morphine	3.667	-1.875	1.088	-1.762	0.960
Heroin	2.167	-3.375	1.088	-3.318	0.960

Question 7

Briefly summarise your conclusions regarding the effects of morphine and heroin on mental activity scores in this trial.

SOLUTION

The F-test from model 2 (obtained using `anova(mod2)`) gives $p = 0.0109$, indicating moderately strong evidence that the groups differ in respect of their post-injection scores. Looking at the p-values corresponding to the individual coefficients, we see that the estimated difference between morphine and placebo is not formally statistically significant at the 5% level, but is close ($p = 0.089$), whilst the estimated difference between heroin and placebo is statistically significant ($p = 0.003$).

Adjusting for ‘prement’ changes the estimated group differences somewhat, but not dramatically. This is as we would expect. Since the trial is randomized there is no confounding. There are only small chance differences between the group means at baseline. Adjusting for these changes the estimated treatment effects slightly, but not materially. The primary purpose of baseline adjustment in randomised trials is to improve the efficiency of treatment effect estimates, this being reflected in (here only slightly) reduced standard errors.

Thus focusing on the effect estimates from model 4, we have that the estimated mean mental activity score in the heroin arm is 3.32 (95% CI 1.40 to 5.23, $p=0.001$) lower than the placebo arm, conditional on (pre)baseline score, which is highly statistically significant. The estimated mean mental activity score in the morphine arm is 1.76 (95% CI -0.15 to 3.68, $p=0.07$) lower than the placebo arm, conditional on (pre)baseline score. There is only weak evidence for the (conditional) mean outcome differing the morphine arm compared to the placebo arm.

The F-test for the interaction parameters in the last model gives $p = 0.43$, indicating no evidence against the null hypothesis that the effects of the drugs are the same irrespective of pre-injection mental activity. So, we cannot say that there is evidence that the effects of the drugs are different for individuals with different levels of pre-injection mental activity.

Question 8

If you have time, explore approaches for obtaining the estimate and confidence interval for the difference in mean post-treatment scores between the heroin and morphine groups in model 4. The coefficient estimates from a model can be extracted using `coef(mod4)`.

SOLUTION

For model for, we adjusted for prement so we are finding the treatment effect conditional on the value of prement. We hence use the `comparisons` function in the `marginalEffects` package.

First we manually construct the difference of interest. The point estimate is given by:

```
coef(mod4)
```

(Intercept)	treatMorphine	treatHeroin	prement
2.8178979	-1.7615096	-3.3182548	0.4539615

```
est <- as.numeric(coef(mod4)[3] - coef(mod4)[2])
est
```

```
[1] -1.556745
```

For the standard error, because the contrast involves two parameters, we need to use the variance covariance matrix estimate of the coefficients, and the fact that $Var(A - B) = Var(A) + Var(B) - 2 \times Cov(A, B)$:

```
#variance covariance matrix of all coefficients is
vcov(mod4)
```

```
      (Intercept) treatMorphine treatHeroin    prement
(Intercept)    0.81979221 -0.475775512 -0.468296888 -0.059828991
treatMorphine -0.47577551    0.922259747  0.461129873  0.002492875
treatHeroin   -0.46829689    0.461129873  0.921792333  0.001246437
prement       -0.05982899    0.002492875  0.001246437  0.009971499
```

```
#variance of 3rd coef - 2nd coef is
vcov(mod4)[3,3] + vcov(mod4)[2,2] - 2*vcov(mod4)[2,3]
```

```
[1] 0.9217923
```

```
#and so SE is
se <- sqrt(vcov(mod4)[3,3] + vcov(mod4)[2,2] - 2*vcov(mod4)[2,3])

#lastly for the 95% CI we need the relevant t-statistic using
#the qt function
est - qt(0.975,df=68)*se
```

```
[1] -3.472595
```

```
est + qt(0.975,df=68)*se
```

```
[1] 0.3591049
```

Doing this calculation manually is a very good way of understanding the statistical calculations required. But in practice we may prefer not to do the calculations manually. One way to achieve the same result is to use the `comparisons` function of the `marginaleffects` package:

```
library(marginaleffects)
comparisons(
  mod4,
  newdata = "mean",
  variables = list(treat = c("Morphine", "Heroin")),
  df="residual"
)
```

Estimate	Std. Error	t	Pr(> t)	S	2.5 %	97.5 %	Df
-1.56	0.96	-1.62	0.11	3.2	-3.47	0.359	68

Term: treat

Type: response

Comparison: Heroin - Morphine

Note that here we are asking **comparisons** to compare mean outcomes at the specified levels of treatment, with the other covariates held at their means. This reduces to the same contrast of coefficients as we are interested in. But it would not if we instead were applying it to model 5 (with interaction), since in this model the difference in means between treatment groups differs according to the pre-treatment value.